



习题课#7 (Oct. 25)

- 1) 求解逆矩阵方法 (解矩阵方程 $AXB=C$)
- 2) 分块初等变换 指导书⁸⁹例1-3讲过. 可讲例9, 12.

△ 矩阵乘积的秩估计. ($A \in F^{m \times n}, B \in F^{n \times l}$).

上一小节已证明 $\text{rank } AB \leq \min(\text{rank } A, \text{rank } B)$, 且

$$\begin{cases} \text{rank } AB = \text{rank } A \Leftrightarrow \ker A + \text{Im } B = F^n \Leftrightarrow \text{Im } B = F^n \Leftrightarrow B \text{ 行满秩.} \\ \text{rank } AB = \text{rank } B \Leftrightarrow \ker A \cap \text{Im } B = 0 \Leftrightarrow \ker A = 0 \Leftrightarrow A \text{ 列满秩.} \end{cases}$$

自然的想法: 是否对 $\text{rank } AB$ 有下界估计?

首先 $\text{rank } AB = 0 \Leftrightarrow AB = 0 \Leftrightarrow \ker A \supseteq \text{Im } B$.

考虑分块矩阵: $n + \text{rank } AB = \text{rank} \begin{pmatrix} I & \\ & AB \end{pmatrix} = \text{rank} \begin{pmatrix} I & \\ A & AB \end{pmatrix}$

$$= \text{rank} \begin{pmatrix} I & -B \\ A & 0 \end{pmatrix} = \text{rank} \begin{pmatrix} A & 0 \\ I & B \end{pmatrix} \geq \text{rank } A + \text{rank } B.$$

即得 Sylvester 秩不等式: $\text{rank } AB \geq \text{rank } A + \text{rank } B - n$.

下面研究何时等号成立 (充要条件). 即何时 $\text{rank } A + \text{rank } B = \text{rank} \begin{pmatrix} A \\ I & B \end{pmatrix}$.

考虑初等行列变换 $P_1 A Q_1 = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$, $P_2 B Q_2 = \begin{pmatrix} I_s & 0 \\ 0 & 0 \end{pmatrix}$, $\text{rank } A = r$, $\text{rank } B = s$.

$$\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} = \begin{pmatrix} P_1 A Q_1 \\ P_2 B Q_2 \end{pmatrix} = \begin{pmatrix} I_r & 0 \\ 0 & I_s & 0 \end{pmatrix}$$

$$\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} = \begin{pmatrix} I_r & 0 \\ 0 & I_s & 0 \end{pmatrix} = \begin{pmatrix} I_r & 0 \\ C_{11} & C_{12} & I_s \\ C_{21} & C_{22} & 0 \end{pmatrix}$$

($s=l$) $\rightarrow \begin{pmatrix} I_r & 0 \\ 0 & 0 & I_s \\ 0 & C_{22} & 0 \end{pmatrix}$ 则等号成立当且仅当 $\text{rank } C_{22} = 0$ 或 $C_{22} = 0$.



$$\text{rank } AB = \text{rank } A + \text{rank } B - n$$

$$\text{rank } AB = \text{rank } A + \text{rank } B - n \quad \text{iff. } \ker A \subseteq \text{Im } B.$$



△ 矩阵加法, 拼接的秩估计.

Prop. 设 $A \in \mathbb{F}^{m \times n}$, $\text{rank} A = r$. 记 $A = P \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} Q^T =: (P \ B) \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} Q_1^T \\ Q_2^T \end{pmatrix}$

其中 P, Q 可逆. 那么易知 $A = P_1 Q_1^T$. 即

$\begin{cases} \text{Im} A = \text{Im} P_1 Q_1^T = \text{Im} P_1, & A \text{ 列空间与 } P_1 \text{ 张成的空间相同.} \\ \text{Im} A^T = \text{Im} Q_1 P_1^T = \text{Im} Q_1, & A \text{ 行空间与 } Q_1 \text{ 张成的空间相同.} \end{cases}$

Proof. 这可直接通过 Q_1^T, P_1^T 行满秩得到, 或根据 P_1, Q_1 列无关, 分别线性表出 A, A^T 列空间而秩相等.

Prop. 上述命题的逆命题在下列意义也成立:

设 $A \in \mathbb{F}^{m \times n}$, $\text{rank} A = r$. $A = P \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} Q^T =: (P_1 \ P_2) \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} Q_1^T \\ Q_2^T \end{pmatrix}$

其中 P, Q 可逆. 则

$\begin{cases} \text{Im} A = \text{Im} P_1 \Rightarrow A_{21} = 0, A_{22} = 0 \\ \text{Im} A^T = \text{Im} Q_1 \Rightarrow A_{12} = 0, A_{22} = 0. \end{cases}$

Proof. $\text{Im} A = \text{Im} (P_1 A_{11} + P_2 A_{21}, P_1 A_{12} + P_2 A_{22}) Q^T = \text{Im} (P_1 A_{11} + P_2 A_{21}, P_1 A_{12} + P_2 A_{22})$
 $= \text{Im} (P_1 A_{11} + P_2 A_{21}) + \text{Im} (P_1 A_{12} + P_2 A_{22})$

$\text{Im} A = \text{Im} P_1 \Rightarrow \text{Im} (P_1 A_{11} + P_2 A_{21}) \subseteq \text{Im} P_1, \text{Im} (P_1 A_{12} + P_2 A_{22}) \subseteq \text{Im} P_1$

$\forall x \exists y \ P_1 A_{11} x + P_2 A_{21} x = P_1 y \quad \exists z \ P_1 A_{12} x + P_2 A_{22} x = P_1 z$

$\Rightarrow \forall x \ A_{21} x = 0, A_{22} x = 0 \quad (P_1 \text{ 与 } P_2 \text{ 列线性无关})$

$\Rightarrow A_{21} = 0, A_{22} = 0.$

$\text{Im} A^T = \text{Im} Q_1$ 时, $A^T = Q \begin{pmatrix} A_{11}^T & A_{21}^T \\ A_{12}^T & A_{22}^T \end{pmatrix} P^T$.

由前述结论, $A_{12}^T = 0, A_{22}^T = 0$, 即 $A_{12} = 0, A_{22} = 0.$



Lem. 令 $A, B \in \mathbb{F}^{m \times n}$. 有 $\text{rank}(A+B) \leq \text{rank} A + \text{rank} B$

令 $A \in \mathbb{F}^{s \times p}, B \in \mathbb{F}^{s \times q}$. 有 $\text{rank}(A, B) \leq \text{rank} A + \text{rank} B$.

Proof. 由于 $\text{Im}(A+B) \subseteq \text{Im} A + \text{Im} B$, 有

$$\begin{aligned} \text{rank}(A+B) &= \dim \text{Im}(A+B) \leq \dim(\text{Im} A + \text{Im} B) = \text{rank}(A, B) \\ &\leq \dim \text{Im} A + \dim \text{Im} B = \text{rank} A + \text{rank} B. \end{aligned}$$

Corl. $\text{rank}(A+B) = \text{rank} A + \text{rank} B \Rightarrow \text{rank}(A, B) = \text{rank} A + \text{rank} B$
for A, B with the same shape.

Thm. For $A, B \in \mathbb{F}^{m \times n}$, TFAE:

a) $\text{rank}(A+B) = \text{rank} A + \text{rank} B$

b) $\text{rank}(A, B) = \text{rank} A + \text{rank} B = \text{rank}(A^T, B^T)$

c) $A = P \begin{pmatrix} \tilde{A} & & \\ & 0 & \\ & & 0 \end{pmatrix} Q^T, B = P \begin{pmatrix} 0 & \tilde{B} & \\ & & \\ & & 0 \end{pmatrix} Q^T$

for non-singular P, Q

Proof. a) $\Rightarrow$ b) Clear by the previous corollary.

b) $\Rightarrow$ c) $\text{rank}(A, B) = \text{rank} A + \text{rank} B \Leftrightarrow \text{Im} A \cap \text{Im} B = \{0\}$.

$$\text{rank}(A^T, B^T) = \text{rank} A^T + \text{rank} B^T \Leftrightarrow \text{Im} A^T \cap \text{Im} B^T = \{0\}.$$

Let $P = (P_1, P_2, P_3), Q = (Q_1, Q_2, Q_3)$ be

$$\text{non-singular matrices s.t. } \begin{cases} \text{Im} A = P_1, \text{Im} B = P_2, \\ \text{Im} A^T = Q_1, \text{Im} B^T = Q_2, \end{cases}$$

then c) follows immediately by the previous proposition.

c) $\Rightarrow$ a) Trivial.



1) (扰动法) 令 $A, B \in F^{n \times n}$. 证明 $(AB)^* = B^* A^*$.

Proof. (直接法) $(AB)^*(i, j) = (-1)^{i+j} (AB) \begin{pmatrix} 1, \dots, j-1, j+1, \dots, n \\ 1, \dots, i-1, i+1, \dots, n \end{pmatrix}$

$$= (-1)^{i+j} \sum_{v_1, \dots, v_n} A \begin{pmatrix} 1, \dots, j-1, j+1, \dots, n \\ v_1, v_2, \dots, v_n \end{pmatrix} B \begin{pmatrix} v_1, v_2, \dots, v_n \\ 1, \dots, i-1, i+1, \dots, n \end{pmatrix}$$

$$= (-1)^{i+j} \sum_{k=1}^n A \begin{pmatrix} 1, \dots, \hat{j}, \dots, n \\ 1, \dots, \hat{k}, \dots, n \end{pmatrix} B \begin{pmatrix} 1, \dots, \hat{k}, \dots, n \\ 1, \dots, \hat{i}, \dots, n \end{pmatrix}$$

$$= (-1)^{i+j} \sum_k (-1)^{j+k} A^*(k, j) (-1)^{k+i} B^*(i, k) = B^* A^*(i, j).$$

(扰动/扰动法) 若 $\det A \neq 0, \det B \neq 0$, 则有

$$(AB)^* = (\det AB)(AB)^{-1} = (\det B) B^{-1} (\det A) A^{-1} = B^* A^*.$$

一般地, 令 $A_\lambda = A + \lambda I, B_\lambda = B + \lambda I$,

至多有 $2n$ 个不同的 λ 使得 $\det A_\lambda = 0$ 或 $\det B_\lambda = 0$.

于是存在 $\delta > 0$ 使得 $\det A_\lambda \neq 0$ 且 $\det B_\lambda \neq 0, \forall 0 < |\lambda| < \delta$.

利用前述结论: $(A_\lambda B_\lambda)^* = B_\lambda^* A_\lambda^*, \forall 0 < |\lambda| < \delta$.

令 $|\lambda| \rightarrow 0$ 即有 $(AB)^* = B^* A^*$.

2) (利用线性方程组得到逆矩阵) 求 $A = \text{diag}(a_1, \dots, a_n) + 11^T, a_i \neq 0, \forall i$.

Proof. 考虑方程组 $Ax = b, x = (x_1, \dots, x_n)^T, b = (b_1, \dots, b_n)^T$.

$$Ax = b \text{ iff } x_1 + \dots + x_n + a_i x_i = b_i, \forall i.$$

将 n 个式子除以 a_i 后相加: $\sum a_i^{-1} (x_1 + \dots + x_n) + (x_1 + \dots + x_n) = \sum b_i / a_i$.

$$\text{即 } x_1 + \dots + x_n = \left(\sum b_i / a_i \right) / \left(1 + \sum a_i^{-1} \right)$$

$$x_j = b_j / a_j - a_j^{-1} (x_1 + \dots + x_n)$$

$$= \frac{b_j}{a_j} - \frac{1}{a_j} \frac{\sum a_i^{-1} b_i}{1 + \sum a_i^{-1}}.$$

令 $A^{-1} = (\beta_1, \dots, \beta_n)$, 那么有 $A\beta_k = e_k$ 为单位向量

取 $b = e_k$, 有 $b_j = \delta_{jk}$, 于是 $x_j = \delta_{jk} a_j^{-1} - a_j^{-1} \frac{a_k^{-1}}{1 + \sum a_i^{-1}}$

$$\begin{aligned} \text{即 } x_j &= \frac{1}{1 + \sum_i a_{ij}} \left[(1 + \sum_i a_{ij}) \delta_{jk} a_{ij}^{-1} - a_{ij}^{-1} a_{ik}^{-1} \right] \\ &= \begin{cases} \frac{1}{1 + \sum_i a_{ij}} ((1 + \sum_i a_{ij}) a_{ik}^{-1} - 1), & j=k, \\ \frac{1}{1 + \sum_i a_{ij}} \left(-\frac{1}{a_{ij} a_{ik}} \right), & j \neq k. \end{cases} \end{aligned}$$

故有 $A^{-1} = -\frac{1}{1+\sum_i C_i^{-1}} (G_j^{-1} G_k^{-1})_{jk} + \text{diag}(G_1^{-1}, \dots, G_k^{-1})$.

△是否等价于矩阵变换?

[illegible]



3) 令 $H = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathbb{R}^{(m+n) \times (m+n)}$, $A \in \mathbb{R}^{m \times m}$, $D \in \mathbb{R}^{n \times n}$. 则 A, D 可逆时
 H 可逆 $\Leftrightarrow A - BD^{-1}C, D - CA^{-1}B$ 均可逆.

Proof. 对 H 考虑分块初等变换:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \rightarrow \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix} \quad \text{乘 } -CA^{-1}$$

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \rightarrow \begin{pmatrix} A - BD^{-1}C & B \\ 0 & D \end{pmatrix}.$$

4) 设 $A \in \mathbb{R}^{n \times n}$. 非对角元非正, 顺序主子式全为正. 证明

a) A^{-1} 存在, 且所有元素非负; b) 存在 $x \in \mathbb{R}^n$ 每个分量为正, Ax 每个分量为正.

Proof. 对阶数归纳. 当 $n=1$ 时显然成立. 假设 a), b) 对 $(n-1)$ 成立

$$A = \begin{pmatrix} A_{11} & \alpha \\ \beta^T & a_{nn} \end{pmatrix} \rightarrow \begin{pmatrix} A_{11} & \alpha \\ 0 & a_{nn} - \beta^T A_{11}^{-1} \alpha \end{pmatrix}$$

$$\text{即 } A = \begin{pmatrix} I & 0 \\ \beta^T A_{11}^{-1} & 1 \end{pmatrix} \begin{pmatrix} A_{11} & \alpha \\ 0 & a_{nn} - \beta^T A_{11}^{-1} \alpha \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} A_{11}^{-1} & -(a_{nn} - \beta^T A_{11}^{-1} \alpha)^{-1} A_{11}^{-1} \alpha \\ 0 & (a_{nn} - \beta^T A_{11}^{-1} \alpha)^{-1} \end{pmatrix} \begin{pmatrix} I & (\text{简单}) \text{三角矩阵的逆} \\ -\beta^T A_{11}^{-1} & 1 \end{pmatrix} \quad \text{熟练掌握.}$$

$$= \begin{pmatrix} A_{11}^{-1} + (a_{nn} - \beta^T A_{11}^{-1} \alpha)^{-1} A_{11}^{-1} \alpha \beta^T A_{11}^{-1} & -(a_{nn} - \beta^T A_{11}^{-1} \alpha)^{-1} A_{11}^{-1} \alpha \\ -(a_{nn} - \beta^T A_{11}^{-1} \alpha)^{-1} \beta^T A_{11}^{-1} & (a_{nn} - \beta^T A_{11}^{-1} \alpha)^{-1} \end{pmatrix}$$

又 $a_{nn} - \beta^T A_{11}^{-1} \alpha = (\det A) / (\det A_{11}) > 0$. α, β 所有元素非正,

于是 $A_{11}^{-1} + (a_{nn} - \beta^T A_{11}^{-1} \alpha)^{-1} A_{11}^{-1} \alpha \beta^T A_{11}^{-1}$ 所有元素非负.

$$-(a_{nn} - \beta^T A_{11}^{-1} \alpha)^{-1} A_{11}^{-1} \alpha, -(a_{nn} - \beta^T A_{11}^{-1} \alpha)^{-1} \beta^T A_{11}^{-1} \geq 0.$$

取 $x = A^{-1} \mathbf{1}$, 则 x 每个分量为正, 且 $Ax = A A^{-1} \mathbf{1} = \mathbf{1}$.



5) (Inverse of Vandermonde matrices) 对于 $W = \det(x_j^i)_{i,j=0}^{n-1}$, 讨论 W^{-1} .

Proof. 只要求得 $W\alpha = e_k, \forall k$ 即可. 不妨 $x_0, \dots, x_{n-1}$ 互不相同

$$\text{即: } \sum_{i=0}^{n-1} x_j^i \alpha_i = \delta_{jk} = \begin{cases} 1, & j=k, \\ 0, & j \neq k. \end{cases}$$

$$\text{令 } f(x) = \sum_{i=0}^{n-1} \alpha_i x^i, \text{ 那么 } f(x_j) = \delta_{jk}.$$

~~* 利用 Lagrange 插值 $f(x)$~~

由于 f 次数至多为 $n-1$, 且已确定了 $x_0, \dots, x_{k-1}, x_{k+1}, \dots, x_{n-1}$ 为零点,

$$\text{可设 } f = C(x-x_0) \cdots (x-x_{k-1})(x-x_{k+1}) \cdots (x-x_{n-1})$$

$$\text{从而进一步得到 } f(x) = \prod_{m \neq k} (x-x_m) / (x_k-x_m).$$

$$\text{得 } \alpha_k \text{ 为 } f \text{ 对应 } x^k \text{ 前系数, 为 } \sum_{\substack{0 \leq i_1 < \dots < i_p < n \\ i_l \neq k}} \prod_{l=1}^p (-x_{i_l}) \prod_{m \neq k} (x_k - x_m)^{-1}.$$

$$\text{故 } W^{-1}(q, r) = W^{-T}(r, q)$$

$$= \sum_{\substack{0 \leq i_1 < \dots < i_r < n \\ i_l \neq k}} (-1)^r \prod_{l=1}^r x_{i_l} \prod_{m \neq k} (x_k - x_m)^{-1}$$

$$= (-1)^r \sigma_r^{(k)} \prod_{m \neq k} (x_k - x_m)^{-1}.$$

$$\text{其中 } \{\sigma_r\}_{r=0}^{n-1} \text{ 满足 } \prod_{m \neq k} (x - x_m) = \sum_{r=0}^{n-1} (-1)^r \sigma_r^{(k)} x^r.$$

特别地, 对于 $Y = V(s^0, \dots, s^{n-1}), s = e^{\frac{2\pi i}{n}}$, 有

$$Y(s, t) = s^{st} \text{ for } s, t = 0, 1, \dots, n-1.$$

$$Y^{-1}(s, t) = (-1)^t \sigma_t^{(s)} \prod_{m \neq s} (s^s - s^m), \quad s, t = 0, 1, \dots, n-1.$$

$$\text{而由 } x^n - 1 = \prod_{m \neq s} (x - s^m) = \sum_{r=0}^{n-1} (-1)^r \sigma_r x^r \quad \& \quad \sigma_t = 0, t=1, \dots, n-2, \\ (-1)^{n-1} \sigma_{n-1} = 1, \sigma_0 = -1.$$



其中 $(-1)^t O_t(s)$ 为 $\prod_{m \neq s} (x - s^m)$ 的 x^t 前系数.

$$\prod_{m \neq s} (x - s^m) = \frac{\prod_{m=0}^{n-1} (x - s^m)}{x - s^s} = \frac{x^n - 1}{x - s^s} = \frac{x^n - s^{sn}}{x - s^s}$$

$$= x^{n-1} + s^s x^{n-2} + s^{2s} x^{n-3} + \dots + s^{(n-1)s}$$

于是 $(-1)^t O_t(s) = s^{(n-1-t)s}$

$$Y^{-1}(s, t) = (-1)^t O_t(s) \prod_{m \neq s} (s^s - s^m)^{-1}$$

$$= s^{(n-1-t)s} \cdot (n s^{(n-1)s})^{-1} = n^{-1} s^{-st}$$

即 $(s^{kl})_{k, l=0}^{n-1}$ 与 $\frac{1}{n} (s^{-st})_{s, t=0}^{n-1}$ 互为逆矩阵.

6) (低秩矩阵更新的逆与行列式) 注意矩阵尺寸

通过分块矩阵变换: $\begin{pmatrix} I_m & B \\ A & I_n \end{pmatrix} \rightarrow \begin{pmatrix} I_m & B \\ 0 & I_n - AB \end{pmatrix}, \begin{pmatrix} I & B \\ A & I \end{pmatrix} \rightarrow \begin{pmatrix} I_n - BA & B \\ 0 & I_n \end{pmatrix}$

易知 $(I - AB)$ 与 $(I - BA)$ 满足

$$m + \text{rank}(I_n - AB) \stackrel{n+}{=} \text{rank}(I_m - BA); \det(I - AB) = \det(I - BA)$$

此时对 A, B 的尺寸无特殊要求, 所以对向量 $\alpha, \beta \in \mathbb{F}^n$ 也有

$$1 + \text{rank}(I_n - \alpha \beta^T) \stackrel{n+}{=} \text{rank}(I - \alpha^T \beta) \leq n+1 \quad \text{等号成立当且仅当 } \alpha^T \beta \neq 1$$

即 $\text{rank}(I_n - \alpha \beta^T) = n$ iff. $\alpha^T \beta \neq 1$.

同时有 $\det(I_n - \alpha \beta^T) = 1 - \alpha^T \beta$.

一般地, $\det(\lambda I_n - AB) = \lambda^n \det(I_n - (\lambda^{-1} A) B) \quad (\lambda \neq 0)$

$$= \lambda^n \det(I_m - B(\lambda^{-1} A)) = \lambda^{n-m} \det(\lambda I_m - BA)$$

即 $\lambda^m \det(\lambda I_n - AB) = \lambda^n \det(\lambda I_m - BA)$. 令 $\lambda \rightarrow 0$ 得此式对所有 λ 成立.

* 直接推论: $\det(I_n - \alpha \beta^T) = 1 - \alpha^T \beta$,

$\Rightarrow \text{rank}(\alpha \beta^T) = 1$. 特征值为 0 (重数 $\overset{\text{至少}}{n-1}$) 和 $\alpha^T \beta$.



在一些场景下我们需要计算形如 $(I_n - A^{-1}B^{-1})$ 的逆矩阵。

级数展开启发: $(1-ab)^{-1} = 1 + ab + (ab)^2 + (ab)^3 + \dots, \forall |ab| < 1$

形式上, $(I - AB)^{-1} = I + AB + (AB)^2 + (AB)^3 + \dots$

$$= I + A(I + BA + (BA)^2 + \dots)B$$

$$= I + A(I - BA)^{-1}B$$

而 $(I - AB)$ 可逆确实能保证 $(I - BA)^{-1}$ 的存在性, 于是

猜想 $(I - AB)^{-1} = I + A(I - BA)^{-1}B$.

直接验证 $(I - AB)(I + A(I - BA)^{-1}B) = I - AB + A(I - BA)^{-1}B - ABA(I - BA)^{-1}B$

$$= I - AB + A(I - BA)(I - BA)^{-1}B = I \text{ 猜想成立.}$$

一般地, 对于可逆 A , 若 V, W 满足 $A - VW$ 可逆, 那么

$$(A - VW)^{-1} = A^{-1}(I - VWA^{-1})^{-1}$$

$$= A^{-1}(I + V(I - WA^{-1}V)^{-1}WA^{-1})$$

$$= A^{-1} + A^{-1}V(I - WA^{-1}V)^{-1}WA^{-1}.$$

此为 Woodbury matrix identity:

$$(A \pm UB)^{-1} = A^{-1} \mp A^{-1}UB(B \pm BVA^{-1}UB)^{-1}BVA^{-1}$$

的特殊情形.

若 V, W 为列/行向量, 有

$$(A \pm \alpha\beta^T)^{-1} = A^{-1} \mp A^{-1}\alpha(1 \pm \beta^TA^{-1}\alpha)^{-1}\beta^TA^{-1} = A^{-1} \mp \frac{A^{-1}\alpha\beta^TA^{-1}}{1 \pm \beta^TA^{-1}\alpha}$$

此为 Sherman-Morrison 公式. (秩 1 更新的逆).

$$\det(A \pm \alpha\beta^T) = \det A \det(I \pm A^{-1}\alpha\beta^T) = \det A (1 \pm \beta^TA^{-1}\alpha) \quad (\text{秩 1 更新的新行列式}).$$

($\det A \neq 0$).

思考: 秩 2 更新行列式应当如何计算?

$$\det(I + ab^T + cd^T) = (1 + a^Tb)(1 + c^Td) - (a^Td)(b^Tc).$$



补充: DFT与循环矩阵. 固定 n 为一正整数. $\zeta = e^{\frac{2\pi i}{n}}$ 为 n 次单位根.

定义如下离散 Fourier 变换: $x \in \mathbb{C}^n \mapsto w \in \mathbb{C}^n$:

$$w_k = \sum_{p=0}^{n-1} x_p e^{-\frac{2\pi i}{n} \cdot pk} \quad k=0, \dots, n-1.$$

容易验证上述变换可表述为矩阵乘积的形式, 即存在 $A \in \mathbb{C}^{n \times n}$, $Ax = w$.

这样的 A 即为 $\zeta^0, \zeta^1, \dots, \zeta^{(n-1)}$ 构成的 Vandermonde 矩阵.

由前述结论知 A 可逆, 且逆矩阵的 n 倍也为 $\zeta^0, \dots, \zeta^{(n-1)}$ 构成的 V -矩阵.

$$\text{DFT: } x \mapsto w = Ax$$

$$\text{IDFT: } A^{-1}w = x \longleftarrow w$$

性质 1: DFT 可将循环矩阵对角化. 令 $C = \begin{pmatrix} c_0 & c_1 & \dots & c_{n-1} \\ c_{n-1} & c_0 & \dots & c_{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ c_1 & c_{n-1} & \dots & c_0 \end{pmatrix} \Rightarrow (\alpha_0, \dots, \alpha_{n-1})$

$$\text{有 } A\alpha_0 = c_0 \zeta^0 + c_{n-1} \zeta^{n-1} + c_{n-2} \zeta^{n-2} + \dots + c_1 \zeta^{n-1}$$

$$A\alpha_1 = c_n \zeta^0 + c_0 \zeta^1 + c_{n-1} \zeta^{-2} + \dots + c_2 \zeta^{(n-1)}$$

$$A\alpha_{n-1} = c_{n-1} \zeta^0 + c_{n-2} \zeta^1 + \dots + c_0 \zeta^{(n-1)}$$

$$\text{容易看出 } A\alpha_k = \zeta^{-k} A\alpha_0, \dots, A\alpha_{n-1} = \zeta^{-1} A\alpha_0 \quad \forall k.$$

$$\text{记 } f(x) = c_0 x^{n-1} + c_1 x^{n-2} + \dots + c_{n-2} x + c_{n-1}$$

$$A\alpha_{n-1} = f(\zeta^{-1}), A\alpha_{n-2} = \zeta f(\zeta^{-1}), \dots, A$$

$$\text{令 } \alpha_k = (1, \zeta^k, \dots, \zeta^{(n-1)k})^T, \text{ 有 } C\alpha_k = f(\zeta^k)\alpha_k.$$

其中 $f(x) = c_0 + c_1 x + \dots + c_{n-1} x^{n-1}$. 于是有

$$nC A^{-1} = nA^{-1} \text{diag}(f(\zeta^0), \dots, f(\zeta^{n-1})) \Rightarrow C = A^{-1} \text{diag}(f(\zeta^0), \dots, f(\zeta^{n-1})) A.$$

$$\text{即: } \begin{array}{ccc} \mathbb{C}^n & \xrightarrow{C} & \mathbb{C}^n \\ \text{DFT} \downarrow & \curvearrowright & \uparrow \text{IDFT} \\ \mathbb{C}^n & \xrightarrow{\text{diag}(f(\zeta^0), \dots, f(\zeta^{n-1}))} & \mathbb{C}^n \end{array}$$



Corl. 循环矩阵的乘积必为循环矩阵.

Proof. 令循环矩阵 C, D 对应的多项式分别为 f 和 g . 则

$$CD = A^{-1} \text{diag}(f(s^0), \dots, f(s^{n-1})) A A^{-1} \text{diag}(g(s^0), \dots, g(s^{n-1})) A \\ = A^{-1} \text{diag}(h(s^0), \dots, h(s^{n-1})) A. \text{ 其中 } h = f \cdot g.$$

对 h 作带余除法: $h = (x^n - 1)q(x) + \tilde{h}$, 那么 $h(s^k) = \tilde{h}(s^k), \forall k$.

$CD = A^{-1} \text{diag}(\tilde{h}(s^0), \dots, \tilde{h}(s^{n-1})) A$ 为循环矩阵, 元素与 $\tilde{h}$ 各项系数一致.

也可利用下述引理说明:

Lem. 令 $\bar{D} = \text{diag}(d_0, \dots, d_{n-1}) \in \mathbb{F}^{n \times 1}$, 则 $A^{-1} \bar{D} A$ 为循环矩阵.

只需找到合适的 $\tilde{h}$ 次数小于 n , 满足 $\tilde{h}(s^k) = d_k$ 即可.

$$\text{令 } \tilde{h} = b_0 x^0 + b_1 x^1 + \dots + b_{n-1} x^{n-1}, \quad \beta := (b_0, \dots, b_{n-1})^T$$

$$\tilde{h}(s^k) = d_k, \forall k \Leftrightarrow A\beta = (d_0, \dots, d_{n-1})^T. \text{ 由 } A \text{ 可逆立刻有 } \beta \text{ 的存在性.}$$

Corl. 循环矩阵的行列式为 $\prod_{k=0}^{n-1} f(s^k)$.

$$\text{Proof. } \det C = \det A^{-1} \det \text{diag}(f(s^0), \dots, f(s^{n-1})) \det A = \prod_{k=0}^{n-1} f(s^k).$$

性质 2: DFT 可将周期边界条件的卷积算子对角化.

固定 $g = (g_{-1}, \dots, g_0, \dots, g_{s-1})^T \in \mathbb{C}^{s+r-1}$. 定义

$$(g \otimes x)_k = \sum_{i: k-i \equiv 0 \pmod{n}} g_i x_i. \text{ 其中 } n \text{ 为 } x \text{ 的长度 (维数).}$$

例如 $g = (g_{-1} = -1, g_0 = 0, g_1 = 1)^T$. $g \otimes x$ 相当于对 x 沿坐标方向导数的二阶差分. $g = (g_{-1} = +1, g_0 = -2, g_1 = 1)^T$ 对应离散的二阶导数. 显然依照定义,

$$g \otimes x = \begin{pmatrix} g_0 & \dots & g_s & 0 & \dots & 0 & g_{s+1} & \dots & g_{n-1} \\ g_1 & & & & & & & & \\ \vdots & & & & & & & & \\ g_{n-1} & & & & & & & & \end{pmatrix} x, \text{ 为循环矩阵对 } x \text{ 的作用.}$$



性质3: 对于(1)DFT矩阵, 计算矩阵向量乘法仅需 $O(n \log n)$ 次操作.
这是因为当 n 是偶数时 $A = \begin{pmatrix} A_0 & A_1 \end{pmatrix}$ 可利用分治的想法递归求解.

(通常称为快速 Fourier 变换 (FFT)).

应用实例: 对两个大整数相乘, 设大整数约为 $O(10^N)$, 则通常的乘法

操作需要 $O(N^2)$ 次操作, 利用 FFT 可加速至 $O(N \log N)$.

类似的思想可推广至有限域 $\mathbb{F}_p$ 中, 取 ω 为原根也有相应的结论.

这一方法通常称为快速数论变换 (NTT, number-theoretic transform).

在处理仅涉及整数的运算时相较 FFT 而言不会受到浮点误差的影响.